

Package ‘bulknul’

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Title Assessing Whether a Bulk-Null Result Can Distinguish Cell-State Dilution from Genuine Absence

Version 0.1.0

Description Diagnostics for bulk transcriptomic null results in studies that also have single-cell data. Computes a Dilution Discordance Score, the posterior probability that a non-significant bulk effect reflects dilution of a small subcluster rather than genuine absence, together with a Diagnosability Index quantifying whether the study design has power to make that distinction at all. Includes an applicability gate that refuses bulk-significant inputs.

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bulknull

One-Call Wrapper for Bulk-Null Dilution Diagnostics

Description

Unified interface that runs the applicability gate, computes the Dilution Discordance Score (DDS), and the Diagnosability Index (DI) in a single call. Returns a formatted S3 object summarizing the complete analysis.

Usage

```
bulknull(
  bulk_beta,
  bulk_se,
  bulk_fdr,
  d_sc = NULL,
  sc_zscore = NULL,
  cluster_fraction,
  condition_fraction,
  r = 1,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass"),
  alpha = 0.05,
  sided = c("one", "two"),
  verbose = FALSE
)
```

Arguments

bulk_beta	numeric; effect size estimate from bulk analysis
bulk_se	numeric; standard error of bulk_beta
bulk_fdr	numeric; FDR (or p-value) from bulk test
d_sc	numeric; standardized effect size in single-cell cluster (primary input)
sc_zscore	numeric; z-score of signal in target cluster (backward compatibility)
cluster_fraction	numeric; fraction of total cells in target cluster
condition_fraction	numeric; fraction of cells in case condition

<code>r</code>	numeric; relative expression (mean in condition / mean across compartment). Default 1 (no relative expression adjustment).
<code>n_cluster</code>	numeric; number of cells in target cluster (required if <code>sc_zscore</code> supplied)
<code>n_eff_basis</code>	character; basis for <code>sc_zscore</code> conversion: "cell" or "donor"
<code>n_donors</code>	numeric; number of donors (required if <code>n_eff_basis</code> = "donor")
<code>null_fdr_threshold</code>	numeric; FDR threshold for applicability gate (default 0.05)
<code>on_violation</code>	character; "error", "warn", or "pass" (default "error")
<code>alpha</code>	numeric; significance level for DI (default 0.05)
<code>sided</code>	character; "one" or "two" for DI computation (default "one")
<code>verbose</code>	logical; if TRUE, include <code>type_s</code> , <code>type_m</code> , and detailed DDS. Default FALSE.

Details

The bulknull workflow is:

1. Check applicability: bulk FDR > threshold (default 0.05)
2. If applicable, compute DDS (posterior probability of dilution)
3. Compute DI (power to detect dilution)
4. Optionally compute `type_s` and `type_m` under model uncertainty

The mixture weight $w = (fr)/(fr + (1-f))$ incorporates relative expression. At $r=1$, this reduces to $w=f$, recovering the standard dilution model.

Value

An S3 object of class "bulknull" containing:

<code>mu_dilution</code>	Expected bulk z-score under dilution hypothesis
<code>z_bulk</code>	Observed bulk z-score
<code>w</code>	Mixture weight used
<code>di</code>	Diagnosability Index value
<code>r</code>	Relative expression parameter used
<code>verdict</code>	Interpretive verdict
<code>inputs</code>	List of all input parameters
<code>applicability</code>	List with <code>\$applicable</code> and <code>\$note</code>
<code>type_s</code>	Type S error rate (when <code>verbose=TRUE</code>)
<code>type_m</code>	Type M error ratio (when <code>verbose=TRUE</code>)
<code>dds</code>	DDS value (when <code>verbose=TRUE</code> only)

Examples

```
# Example: DPP9 in M8 (direct d_sc input, recommended)
result <- bulknull(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  d_sc = 0.117962,
```

```

cluster_fraction = 0.0695,
condition_fraction = 0.631,
alpha = 0.05,
sided = "one"
)

print(result)

```

check_dds_applicability

Check DDS Applicability: Bulk Must Be Null

Description

Validates whether a case is applicable for DDS analysis.

DDS answers the question: "Given a bulk-NULL result and strong single-cell signal, is the null due to dilution or true absence of effect?"

This function enforces the applicability boundary:

- **Applicable:** bulk FDR > null_fdr_threshold (non-significant bulk result)
- **Not Applicable:** bulk FDR <= null_fdr_threshold (significant bulk result)

When not applicable, DDS is mathematically valid but answers the wrong question.

Usage

```

check_dds_applicability(
  bulk_fdr,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass")
)

```

Arguments

- | | |
|--------------------|---|
| bulk_fdr | Numeric. FDR-adjusted p-value from bulk analysis. |
| null_fdr_threshold | Numeric. FDR threshold for bulk to be considered "null". Default: 0.05 (standard significance threshold). |
| on_violation | Character. One of: <ul style="list-style-type: none"> • "error": Stop with informative error message (strict mode) • "warn": Issue warning but return results (permissive mode) • "pass": Return results silently (developer/testing mode) |

Value

A list with elements:

- `applicable`: Logical. TRUE if `bulk_fdr > null_fdr_threshold` (NULL), FALSE otherwise.
- `applicability_note`: Character string explaining the status.
- `violation_severity`: Character. "none", "warning", or "error" depending on `on_violation` setting.
- `bulk_fdr`: The input FDR value.
- `null_fdr_threshold`: The threshold used.
- `call`: The matched function call.

Examples

```
# Case 1: Bulk is non-significant (FDR = 0.15) - APPLICABLE
check_dds_applicability(bulk_fdr = 0.15, null_fdr_threshold = 0.05,
                        on_violation = "warn")

# Case 2: Bulk is significant (FDR = 0.001) - NOT APPLICABLE
check_dds_applicability(bulk_fdr = 0.001, null_fdr_threshold = 0.05,
                        on_violation = "warn")
```

cohort_inflation	<i>Cohort Inflation Factor for Target Diagnosability Index</i>
------------------	--

Description

Compute the required inflation in cohort size (and thus reduction in bulk SE) to achieve a target diagnosability index. The inflation factor is independent of the pooled cohort size N and the bulk point estimate; it depends only on the observed and required effect sizes μ .

Usage

```
cohort_inflation(
  mu_observed,
  target_di = 0.8,
  r = 1,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

<code>mu_observed</code>	numeric; observed effect size (bulk z-score / $\sqrt{n_{\text{eff_bulk}}}$)
<code>target_di</code>	numeric; target diagnosability index (0 to 1, default 0.8)
<code>r</code>	numeric; relative expression parameter. Default 1 (no adjustment). Used for reference only.
<code>alpha</code>	numeric; significance level (default 0.05)
<code>sided</code>	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

Identity: $\text{required inflation} = (\mu_{\text{required}} / \mu_{\text{observed}})^2$

The relationship between bulk SE and cohort size is: $\text{SE} \sim 1 / \sqrt{N}$

To reduce SE by a factor k , cohort size must increase by k^2 . Therefore, if the current μ is μ_{obs} and we need μ_{req} , the inflation is: $\text{inflation} = (\mu_{\text{req}} / \mu_{\text{obs}})^2$

This computation is independent of:

- The absolute cohort size
- The bulk point estimate (only μ_{obs} matters)
- The cluster fraction or single-cell effect size

An inflation of 50x means the cohort must grow 50 times in size. An inflation of 50 with current $N=630$ implies new $N \sim 31,500$.

Value

A numeric scalar: the required inflation factor (≥ 1.0 if target is attainable)

Examples

```
inflation <- cohort_inflation(
  mu_observed = 0.130536,
  target_di = 0.8,
  alpha = 0.05,
  sided = "one"
)

cat("Required inflation:", inflation, "\n")
cat("If current N=630, new N would be:", 630 * inflation, "\n")
```

critical_precision	<i>Critical Precision for Target Diagnosability Index</i>
--------------------	---

Description

Compute the minimum bulk measurement precision (SE) required to achieve a target diagnosability index at any cluster fraction ≤ 1 . A target DI is attainable if and only if $\text{bulk_se} \leq \text{s_critical}$, which depends only on d_sc , target_di , and α .

Usage

```
critical_precision(
  d_sc,
  bulk_se,
  target_di = 0.8,
  r = 1,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

d_sc	numeric; standardized effect size in single-cell cluster
bulk_se	numeric; observed bulk measurement standard error
target_di	numeric; target diagnosability index (0 to 1, default 0.8)
r	numeric; relative expression parameter. Default 1 (no adjustment). Used for reference only.
alpha	numeric; significance level (default 0.05)
sided	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

The relationship is independent of the pooled cohort size N ; it depends only on the effect size standardization scale and the statistical threshold.

The condition for attainability is: $\text{bulk_se} \leq d_sc / (z_critical + qnorm(\text{target_di}))$

This formula is independent of N (the cohort size): it depends only on the statistical threshold and effect-size scale.

The minimum detectable effect at $f=1$ is: $\text{min_d} = \text{mu_required} * \text{bulk_se}$

For a hypothesised within-cluster effect d , the required cluster fraction is: $\text{f_required} = \text{mu_required} * \text{bulk_se} / d$

Value

A list with elements:

mu_required	Required effect size mu to achieve target DI
s_critical	Critical bulk SE; target DI attainable iff $\text{bulk_se} \leq s_critical$
attainable	Logical; TRUE if target DI is attainable at current bulk_se
min_detectable_d	Smallest within-cluster effect reachable at $f=1.0$
f_required_at_d	Function $\text{f_required_at_d}(d)$ returning required cluster fraction
call	The matched function call

Examples

```
result <- critical_precision(
  d_sc = 0.117962,
  bulk_se = 0.062774,
  target_di = 0.8,
  alpha = 0.05,
  sided = "one"
)

print(result$attainable) # FALSE: bulk_se exceeds s_critical
print(result$min_detectable_d)
```

dds_bounds

*DDS Bounds Over Bulk Z-Score Range***Description**

Compute the attainable interval of Dilution Discordance Scores (DDS) for a given `mu_dilution` (effect size) as the bulk z-score `z_bulk` varies over a specified range.

Usage

```
dds_bounds(mu, bulk_se = NULL, z_lo = -1.96, z_hi = 1.96)
```

Arguments

<code>mu</code>	numeric; effect size <code>mu</code> under dilution hypothesis
<code>bulk_se</code>	numeric; bulk measurement standard error (used only for <code>min_detectable_d</code>)
<code>z_lo</code>	numeric; lower bound of bulk z-score range (default -1.96)
<code>z_hi</code>	numeric; upper bound of bulk z-score range (default 1.96)

Details

The DDS is a function of both `mu_dilution` and `z_bulk`. For fixed `mu_dilution`, extreme values of `z_bulk` produce extreme DDS values. This function returns the interval of DDS that could arise if the true bulk z-score were anywhere in the specified range (typically between negative 1.96 and positive 1.96 for typical bulk analyses).

Identity: $DDS(z) = \text{plogis}(\mu * z - \mu^2/2)$ where `plogis` is the logistic CDF (inverse of `logit`).

The DDS is computed as: $DDS = \text{plogis}(\mu * z - \mu^2/2)$

For fixed `mu`, as `z` varies from `z_lo` to `z_hi`, the DDS varies accordingly. The function returns the minimum and maximum attainable DDS values.

A wide interval means many DDS values are possible (uncertainty in true `z_bulk`). A narrow interval means few DDS values are possible (strong constraint on outcome).

Value

A list with elements:

<code>lower</code>	Lower bound of attainable DDS interval
<code>upper</code>	Upper bound of attainable DDS interval
<code>width</code>	Width of the interval (<code>upper - lower</code>)
<code>reaches_0.7</code>	Logical; TRUE if interval reaches $DDS \geq 0.7$
<code>reaches_0.9</code>	Logical; TRUE if interval reaches $DDS \geq 0.9$
<code>call</code>	The matched function call

Examples

```

bounds <- dds_bounds(
  mu = 0.130536,
  bulk_se = 0.062774,
  z_lo = -1.96,
  z_hi = 1.96
)

cat("DDS range:", bounds$lower, "to", bounds$upper, "\n")
cat("Interval width:", bounds$width, "\n")
cat("Reaches strong evidence (0.7)?", bounds$reaches_0.7, "\n")

```

diagnosability_index *Diagnosability Index (DI)*

Description

Compute statistical power to detect dilution under a one-sided or two-sided test.

Usage

```
diagnosability_index(mu_dilution, alpha = 0.05, sided = c("one", "two"))
```

Arguments

mu_dilution	Numeric. Expected signal under dilution: $\mu = (\text{cluster_frac} * \text{sc_zscore}) / (\text{bulk_se} * \text{sqrt}(n_{\text{eff}}))$.
alpha	Numeric. Significance level (default 0.05).
sided	Character. "one" for one-sided test (default, recommended for directional alternatives) or "two" for two-sided.

Details

DI estimates the probability of correctly detecting dilution if it truly occurred at the specified effect size (mu_dilution).

One-sided (default, recommended):

- Critical value: $z = \text{qnorm}(1 - \alpha) = 1.644854$ for $\alpha=0.05$
- Rationale: Dilution alternative is directional by construction ($\text{sign}(\mu_{\text{dilution}}) = \text{sign}(\text{sc_zscore})$ since $\text{cluster_frac} > 0$)
- Formula: $\text{DI} = \text{pnorm}(\mu - 1.644854)$

Two-sided (for reference):

- Critical value: $z = \text{qnorm}(1 - \alpha/2) = 1.959964$ for $\alpha=0.05$
- Formula: $\text{DI} = \text{pnorm}(\mu - 1.959964)$

Power categories:

- $\text{DI} < 0.3$: Low power (undetectable at this effect size)
- $\text{DI} 0.3\text{--}0.8$: Moderate power (detection possible but unreliable)
- $\text{DI} > 0.8$: High power (reliable detection)

Value

List containing:

- di: Diagnosability Index (power), in range 0 to 1
- critical_value: Critical z-value used
- alpha: Significance level
- sided: Test type ("one" or "two")
- formula: Human-readable formula string

Examples

```
# DPP9/M8 k=5 tissue-only
diagnosability_index(mu_dilution = 0.130536, alpha = 0.05, sided = "one")
# Returns: di = 0.0650 (very low power; dilution undetectable)
```

dilution_scan	<i>Vectorized Dilution Scan Across Multiple Genes</i>
---------------	---

Description

Applies the bulknull framework to multiple genes in a single call, returning a summary table with DDS, DI, and verdict for each gene. Skips bulk-significant genes ($FDR \leq \text{threshold}$) and genes with missing single-cell entries.

Usage

```
dilution_scan(
  bulk_deg,
  d_sc_values = NULL,
  sc_zscores = NULL,
  cluster_fraction,
  condition_fraction,
  r = 1,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  null_fdr_threshold = 0.05,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

bulk_deg	data frame; differential expression results with columns gene (character), beta or log2fc (numeric effect size), se or se_log2fc (numeric standard error), and fdr or padj (numeric FDR). Additional columns are ignored.
d_sc_values	named numeric vector; standardized effect sizes keyed by gene (primary input)
sc_zscores	named numeric vector; z-scores keyed by gene (backward compatibility). Genes not in either vector are skipped with a note.

cluster_fraction	numeric; fraction of total cells in target cluster (constant across all genes).
condition_fraction	numeric; fraction of cells in case condition (constant across all genes).
r	numeric; relative expression parameter. Default 1.
n_cluster	numeric; number of cells in target cluster (constant across all genes). Required if sc_zscores supplied.
n_eff_basis	character; "cell" or "donor" basis for z-score conversion.
n_donors	numeric; number of donors (required if n_eff_basis = "donor").
null_fdr_threshold	numeric; FDR threshold for applicability gate (default 0.05). Genes with FDR $\leq$ threshold are marked "FAILED".
alpha	numeric; significance level for DI (default 0.05).
sided	character; "one" or "two" for one-sided or two-sided test (default "one").

Details

The function iterates over rows of `bulk_deg`, skipping:

- Genes with bulk FDR $\leq$ `null_fdr_threshold` (gate fails)
- Genes without an entry in `sc_zscores`
- Genes with zero variance in the single-cell data (`sc_zscore` = 0)

For applicable genes, it calls `bulknull()` internally and extracts key results. Returns all processed genes in output, with NA or "SKIPPED" for missing data.

Value

A data frame with columns:

gene	Gene name from <code>bulk_deg</code>
bulk_beta	Effect size from <code>bulk_deg</code>
bulk_se	Standard error from <code>bulk_deg</code>
bulk_fdr	FDR from <code>bulk_deg</code>
sc_zscore	Single-cell z-score (NA if not found)
mu_dilution	Expected bulk z-score under dilution hypothesis
dds	Dilution Discordance Score
di	Diagnosability Index
dds_interpretation	Categorical interpretation of DDS
di_interpretation	Categorical interpretation of DI
verdict	Final interpretive verdict
gate_status	PASSED or FAILED (applicability gate status)

Examples

```
# Example: scan a small DEG table with mock z-scores
bulk_deg <- data.frame(
  gene = c("GENE1", "GENE2", "GENE3"),
  beta = c(-0.084, 0.020, 0.150),
  se = c(0.063, 0.050, 0.100),
  fdr = c(0.450, 0.05, 0.001)
)
sc_zscores <- c(GENE1 = 2.077, GENE2 = 1.5, GENE3 = 0.5)

result <- dilution_scan(
  bulk_deg = bulk_deg,
  sc_zscores = sc_zscores,
  cluster_fraction = 0.0695,
  condition_fraction = 0.631,
  n_cluster = 1332
)
print(result)
```

dilution_score

*Dilution Discordance Score (DDS)***Description**

Compute the Bayesian posterior probability that an observed bulk-level effect reflects true dilution of a single-cell signal.

Usage

```
dilution_score(
  bulk_beta,
  bulk_se,
  bulk_fdr,
  d_sc = NULL,
  sc_zscore = NULL,
  cluster_fraction,
  condition_fraction,
  r = 1,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass")
)
```

Arguments

bulk_beta	Numeric. Observed bulk effect size (Hedges' g or log2-fold-change).
bulk_se	Numeric. Standard error of bulk effect size.
bulk_fdr	Numeric. FDR-adjusted p-value from bulk analysis (or raw p-value). Used for validation only; not directly in likelihood.

d_sc	Numeric. Standardized effect size in the single-cell cluster (primary input). When supplied, sc_zscore conversion is skipped.
sc_zscore	Numeric. Single-cell z-score (backward compatibility). Ignored if d_sc supplied. Must provide either d_sc or sc_zscore, not both unless testing.
cluster_fraction	Numeric. Fraction of total tissue represented by the responsible subcluster ($0 < f \leq 1$).
condition_fraction	Numeric. Fraction of subcluster cells from the condition of interest (e.g., IPF, $0 \leq p \leq 1$).
r	Numeric. Relative expression: mean expression of gene in condition divided by mean expression across whole compartment. Default $r = 1$. The mixture weight $w = (fr)/(fr + (1-f))$ is bounded above by 1, so mu_dilution is bounded by d_sc/bulk_se regardless of r value.
n_cluster	Integer. Number of cells in the subcluster. Required if sc_zscore supplied.
n_eff_basis	Character. Basis for sc_zscore conversion: "cell" or "donor". Default "cell". Ignored if d_sc supplied.
n_donors	Integer. Number of donors. Required if n_eff_basis = "donor".
null_fdr_threshold	Numeric. FDR threshold for bulk to be considered "null". Default: 0.05. Only used if bulk_fdr is provided.
on_violation	Character. One of "error", "warn", "pass". Controls behavior when bulk FDR is significant. Default: "warn".

Value

A list with elements:

- dds_score: Posterior probability of dilution, between 0 and 1.
- z_bulk: Bulk z-score (beta / SE).
- mu_dilution: Expected z under dilution model.
- w: Mixture weight = $(fr)/(fr + (1-f))$, bounded between 0 and 1.
- d_sc: Standardized effect size used (supplied or converted).
- log_lik_h1: Log-likelihood under dilution model.
- log_lik_h0: Log-likelihood under null model.
- summary: Plain-text interpretation.
- applicable: Logical. TRUE if case is applicable (bulk is null), FALSE if not.
- applicability_note: Character string describing applicability status.
- call: The matched function call.

Examples

```
# Direct d_sc input (recommended)
dilution_score(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  d_sc = 0.117962,
```

```

    cluster_fraction = 0.0695,
    condition_fraction = 0.631
)

# Backward compatible: sc_zscore with cell-level basis
dilution_score(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  sc_zscore = 2.077,
  cluster_fraction = 0.0695,
  condition_fraction = 0.631,
  n_cluster = 1332,
  n_eff_basis = "cell"
)

```

dpp9_m8

DPP9/M8 Example Dataset

Description

A list containing measurements for DPP9 (diphosphonucleotidase 9) expression in the M8 (inflammatory macrophages) subcluster from IPF lung tissue.

Usage

```
data(dpp9_m8)
```

Format

A list with 12 elements:

bulk_beta Numeric. Bulk effect size (-0.0297)

bulk_se Numeric. Standard error of bulk effect (0.0595)

bulk_fdr Numeric. FDR-adjusted p-value from meta-analysis (0.821)

sc_zscore Numeric. Single-cell module z-score (2.077)

cluster_fraction Numeric. Fraction of tissue in M8 (0.0695)

condition_fraction Numeric. Fraction of M8 from IPF (0.631)

n_cluster Integer. Number of cells in M8 (1332)

gene Character. Gene name ("DPP9")

cluster Character. Cluster identifier ("M8")

cluster_name Character. Cluster description ("Inflammatory Macrophages")

cohort Character. Disease cohort ("IPF")

description Character. Dataset description

Details

This is a real data example from the bulknull application: a bulk-level null result (non-significant effect) paired with strong single-cell signal in a small subcluster. Used throughout vignettes and examples.

The DPP9/M8 case exemplifies the framework's utility: bulk RNA-seq from IPF lung tissue shows no significant DPP9 effect (FDR = 0.821), but single-cell analysis reveals strong inflammasome-module enrichment in M8. The DDS computes whether the bulk null reflects true absence or dilution of this M8 signal by the dominant other cell types.

Result: DDS ~ 0.48 (ambiguous), suggesting both null and dilution remain plausible. The Diagnosability Index (DI ~ 0.065) reveals that the study design is fundamentally underpowered for dilution detection due to the small M8 fraction (6.95% of lung), independent of effect size.

Examples

```
data(dpp9_m8)
str(dpp9_m8)

# Compute Dilution Discordance Score
result <- dilution_score(
  bulk_beta = dpp9_m8$bulk_beta,
  bulk_se = dpp9_m8$bulk_se,
  bulk_fdr = dpp9_m8$bulk_fdr,
  sc_zscore = dpp9_m8$sc_zscore,
  cluster_fraction = dpp9_m8$cluster_fraction,
  condition_fraction = dpp9_m8$condition_fraction,
  n_cluster = dpp9_m8$n_cluster
)
print(result$dds_score)
```

plot.bulknull

Plot DI Landscape for Study Design

Description

Visualizes the Diagnosability Index as a function of cluster fraction and bulk measurement precision (SE), marking the current study's position. Base graphics only.

Usage

```
## S3 method for class 'bulknull'
plot(
  x,
  sc_zscore = NULL,
  condition_fraction = NULL,
  n_cluster = NULL,
  alpha = 0.05,
  sided = c("one", "two"),
  cluster_fraction_range = c(0.001, 1),
  se_range = c(0.001, 0.5),
  grid_points = 50,
  ...
)
```

Arguments

x	A bulknull object
sc_zscore	numeric; single-cell z-score (same as input to bulknull)
condition_fraction	numeric; case fraction in cohort
n_cluster	numeric; cells in target cluster
alpha	numeric; significance level (default 0.05)
sided	character; "one" or "two" (default "one")
cluster_fraction_range	numeric vector of length 2; range for y-axis
se_range	numeric vector of length 2; range for x-axis (log scale)
grid_points	numeric; number of grid points per dimension (default 50)
...	Additional arguments (ignored)

Details

The landscape shows DI as a 2D function of cluster_fraction and bulk_se. Higher DI (>0.8) indicates good power; lower DI (<0.3) indicates underpowered. Contours mark DI = 0.3 and DI = 0.8 boundaries. The study is marked with a red dot at (bulk_se, cluster_fraction) from the input bulknull object.

print.bulknull

Print method for bulknull objects

Description

Displays a formatted verdict block showing inputs, computed statistics, interpretations, and the final verdict.

Usage

```
## S3 method for class 'bulknull'
print(x, ...)
```

Arguments

x	A bulknull object
...	Additional arguments (ignored)

```
print.critical_precision
```

Print method for critical_precision objects

Description

Print method for critical_precision objects

Usage

```
## S3 method for class 'critical_precision'
print(x, ...)
```

Arguments

x	A critical_precision object
...	Additional arguments (ignored)

```
print.dds_bounds
```

Print method for dds_bounds objects

Description

Print method for dds_bounds objects

Usage

```
## S3 method for class 'dds_bounds'
print(x, ...)
```

Arguments

x	A dds_bounds object
...	Additional arguments (ignored)

```
print.required_design
```

Print method for required_design objects

Description

Print method for required_design objects

Usage

```
## S3 method for class 'required_design'
print(x, ...)
```

Arguments

x	A required_design object
...	Additional arguments (ignored)

required_design

*Required Study Design to Achieve Target Diagnosability Index***Description**

Inverts the diagnosability index formula to find the study parameters needed to achieve a target power (DI). Can solve for either cluster_fraction (at fixed bulk_se) or bulk_se (at fixed cluster_fraction).

Usage

```
required_design(
  target_di,
  d_sc = NULL,
  sc_zscore = NULL,
  condition_fraction,
  r = 1,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  cluster_fraction,
  bulk_se,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

target_di	numeric; target diagnosability index (0 to 1)
d_sc	numeric; standardized effect size in single-cell cluster (primary input)
sc_zscore	numeric; single-cell z-score (backward compatibility)
condition_fraction	numeric; fraction of cells in case condition (0 to 1)
r	numeric; relative expression parameter. Default 1.
n_cluster	numeric; number of cells in target cluster (required if sc_zscore supplied)
n_eff_basis	character; basis for conversion: "cell" or "donor"
n_donors	numeric; number of donors (required if n_eff_basis = "donor")
cluster_fraction	numeric; fraction of total cells in target cluster
bulk_se	numeric; standard error of bulk effect estimate
alpha	numeric; significance level (default 0.05)
sided	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

The diagnosability index is defined as: $DI = \text{Normal_CDF}(\mu - z_{\text{critical}})$ where `Normal_CDF` is the standard normal CDF and `z_critical` depends on `alpha` and `sided`.

This function inverts the formula in two ways:

1. Solves for `cluster_fraction` given `bulk_se`, `sc_zscore`, and `condition_fraction`
2. Solves for `bulk_se` given `cluster_fraction`, `sc_zscore`, and `condition_fraction`

If the required `cluster_fraction` exceeds 1 (unattainable), the function returns NA and includes a warning message.

Value

A list with class "required_design" containing:

<code>target_di</code>	Requested diagnosability index
<code>achieved_di_at_fixed_se</code>	DI achieved if solving for <code>cluster_fraction</code>
<code>required_cluster_fraction</code>	Cluster fraction needed to reach target DI (at fixed <code>bulk_se</code>)
<code>required_bulk_se</code>	Bulk SE needed to reach target DI (at fixed <code>cluster_fraction</code>)
<code>achieved_di_at_fixed_cf</code>	DI achieved if solving for <code>bulk_se</code>
<code>scenario_fixed_se</code>	Message describing the fixed-SE scenario
<code>scenario_fixed_cf</code>	Message describing the fixed-CF scenario
<code>warnings</code>	Character vector of warnings if targets are unattainable

Examples

```
# Example: k=5 tissue-only case study
# Current DI is only 0.065; what would we need for DI = 0.3?

design <- required_design(
  target_di = 0.3,
  sc_zscore = 2.077409,
  condition_fraction = 0.631,
  n_cluster = 1332,
  cluster_fraction = 1332 / 19175,
  bulk_se = 0.062774,
  alpha = 0.05,
  sided = "one"
)

print(design)
```

summary.bulknull

Summary method for bulknull objects

Description

Displays a concise one-line summary of the diagnosis.

Usage

```
## S3 method for class 'bulknull'  
summary(object, ...)
```

Arguments

object	A bulknull object
...	Additional arguments (ignored)